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Paleomagnetism and tectonic significance of the Goble Volcanic Series, southwestern Washington

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ABSTRACT

The mean direction of remanent magnetism of the upper Eocene–Oligocene Goble Volcanic Series, located on the western flank of the Cascades in southwestern Washington, points about 25° east of the expected middle Tertiary geomagnetic field direction for the area. This deflection is in the same sense as discrepant paleomagnetic directions reported for Tertiary rocks from the Oregon Coast Range, but seems to be somewhat less in magnitude. From their major-element chemistry and their structural location, it appears that the Goble flows may be an early arc assemblage that was ancestral to the present Cascade magmatic arc, although origin as an oceanic basalt assemblage akin to the Yachats basalts and the Siletz River and Tillamook Volcanic Series of the Oregon Coast Range cannot be ruled out. Whatever its origin, the Goble block has been rotated about a pivot located somewhere nearby, either as part of the Coast Range or as an independent microcontinental block.

INTRODUCTION

New paleomagnetic data from western Oregon (Simpson and Cox, 1977; Plumley and Beck, 1977; Cox and McGill, 1977) all are consistent with an astounding tectonic model that requires the Oregon Coast Range to have rotated at least 60° clockwise, relative to the North American interior, since early Eocene time. The evidence for this rotation consists of the observation that all pre-Miocene rock units in the Coast Range so far studied paleomagnetically are magnetized in a direction that lies well eastward of the direction required by the known position of the pole, relative to North America, at the time the rock unit in question was formed. One such discordant paleomagnetic direction, of course, would not be very convincing, but in the Coast Range there are several such peculiarly magnetized rock units. These units consist of basalts, bedded clastic sedimentary rocks, and dikes and sills that range in age from early Eocene to Oligocene, all of which show the same sense of discordance. Moreover, the amount of discordance (specifically, the angle between the expected and measured magnetic declinations) decreases with decreasing age of the unit, which suggests that the Coast Range block may have "rotated into place" at the same time that the rocks currently exposed on its upper surface were forming (Simpson and Cox, 1977; Plumley and others, 1978). The consistency of this picture over a block at least 350 km long and 50 km wide, which embraces rocks magnetized by different processes over a time span of about 25 m.y., eliminates the possibility that the discordant directions are the product of local tectonics or peculiarities ("excursions") of the geomagnetic field. Thus, the fact that the Oregon Coast Range has rotated seems well established. The mechanism by which it accomplished its rotation, however, is not well understood.

Because one of the most important constraints on rotation models for the Pacific Northwest is the size of the rotated block, we set out to determine if the Goble Volcanic Series, which is located north and east of known rotated exposures in Oregon, also is rotated. Our results indicate that the Goble exposures have indeed been rotated but that they may be part of an independent crustal block.

GEOLOGIC SETTING

The Goble Volcanic Series was named by Wilkinson and others (1946) for exposures of basaltic flows on the Columbia River near Goble, Oregon. The bulk of the exposure, however, lies east of the river in southwestern Washington (Fig. 1), where the series occupies a position between the younger High Cascade volcanic series and the axis of the Puget-Willamette lowland. The series consists of flows of basaltic to andesitic composition, with some pyroclastic rocks and minor sedimentary rocks of volcaniclastic derivation. The rocks appear to be mainly of subaerial origin. The age of the Goble Volcanic Series, based on fossil assemblages within the Cowlitz Formation with which the Goble volcanic rocks interfinger, is late Eocene to early Oligocene. Whole-rock K-Ar ages (kindly provided to us by R. A. Duncan of Oregon State University) confirm this age designation. Replicate measurements from site 77-07B, located about 29 km east

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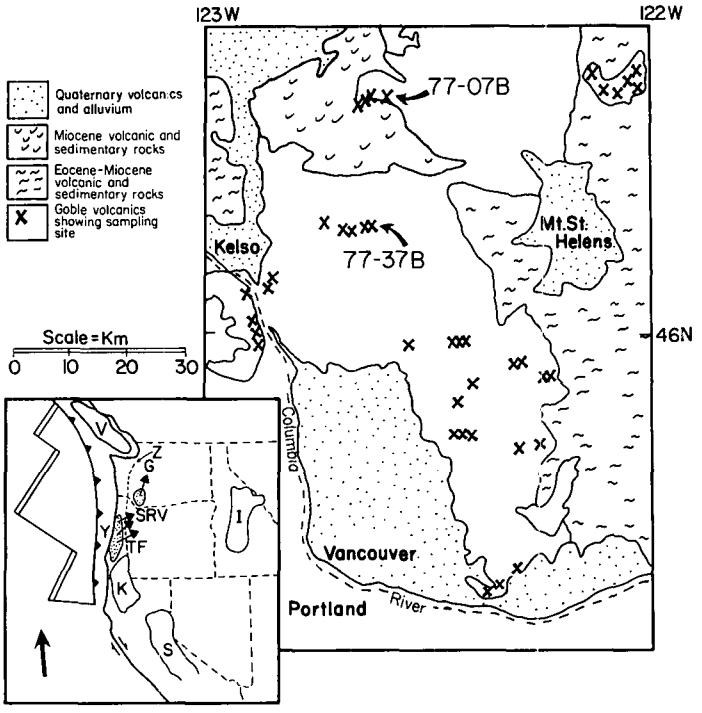


Figure 1. Location map of part of Washington (north of Columbia River) and Oregon (south of Columbia River) showing sampling localities, modified from Huntting and others (1961). Inset, modified from Simpson and Cox (1977), shows tectonic setting of the Oregon Coast Range and the Goble block (dotted). Small arrows show present declination of remanent magnetization for the Goble Volcanic Series (G), Siletz River Volcanic Series (SRV), Yachats basalts (Y), and the Tyee-Flournoy sediments (TF). Large arrow shows Eocene expected declination. I = Idaho batholith; K = Klamath Mountains; S = Sierra Nevada; V = Vancouver Island; Z = Bouguer gravity gradient. Detailed site descriptions in Burr (1978).

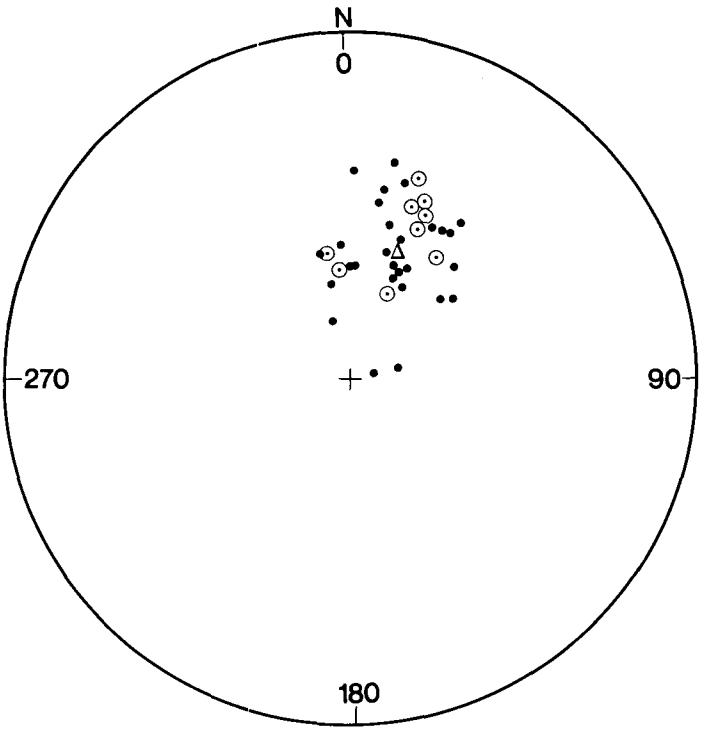


Figure 2. Mean directions of remanent magnetization for individual flows within the Goble Volcanic Series. Lower-hemisphere, equal-area plot. Solid circles are normal polarity; open circles are reverse polarity inverted through the center of the projection. Triangle indicates mean direction.

of Interstate 5 on Washington Highway 504 along the Toutle River, yielded ages of 32.2 ± 0.3 m.y. and 35.9 ± 0.4 m.y. (early Oligocene). A second pair of replicate measurements made on material from site 77-37B, 19 km east of Interstate 5 along the Coweman River Road, gave ages of 45.0 ± 1.4 m.y. and 41.4 ± 1.3 m.y. (late Eocene). The Goble Volcanic Series is at least 1,500 m thick, on the basis of the type section measured by Wilkinson and others (1946) along the Columbia River north of Portland. Flows range in thickness from less than 1 m to 30 m and are typically massive basalt or basaltic andesite lying on a rubbly breccia base that commonly forms irregular surfaces. Dips within the series are difficult to measure but tend to be small (<25°), with most of the flows having dips of 10° or less. Minor faulting was observed at several outcrops within the study area, but major faults that might duplicate the stratigraphic section have not been recognized. Burr (1978) has presented a more complete discussion of the geology and petrology of the Goble Volcanic Series.

PALEOMAGNETISM AND ORIGIN OF THE GOBLE VOLCANIC SERIES

We collected 392 samples from 42 flows distributed mainly along seven major west-flowing river systems traversing the Goble outcrop area (Fig. 1). All samples were magnetically cleaned by

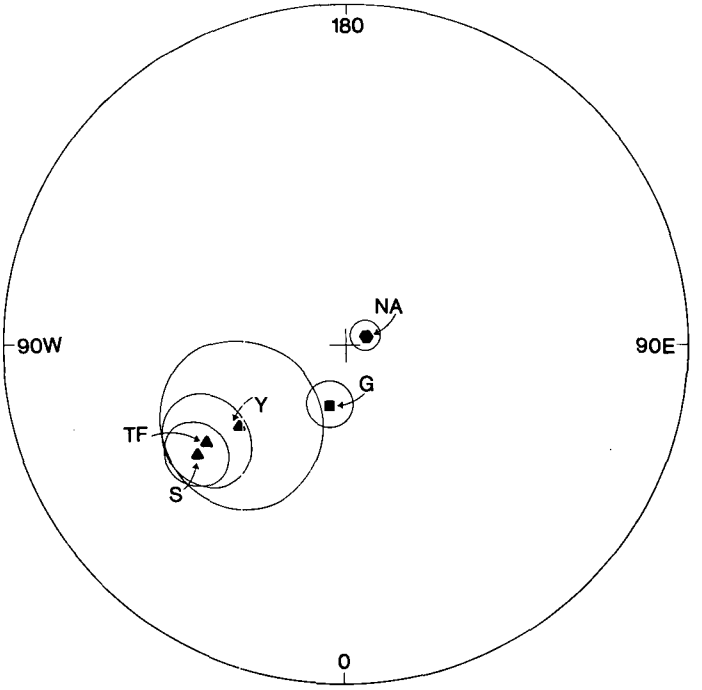


Figure 3. Paleomagnetic poles and 95% confidence intervals for units from the Oregon Coast Range and the Goble Volcanic Series. S = Siletz River Volcanic Series; TF = Tyee-Flournoy sediments; Y = Yachats basalts; G = Goble Volcanic Series. NA = mid-Tertiary reference pole for interior North America.

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TABLE 1. MEASURED AND EXPECTED DIRECTIONS OF MAGNETIZATION AND PALEOMAGNETIC POLES FOR THE GOBLE VOLCANIC SERIES

	D (°)	I (°)	R	α_{95}	PLONG	PLAT	R	α_{95}
Goble Volcanic Series	18.5	57.5	35.82	4.3	14.5W	75.5N	34.97	5.5
Middle Tertiary, stable North America	353.5	61.5			115E	84.5N		3.5

Explanation: D and I are mean declination and inclination, respectively; R is the length of the vector sum derived in the calculation of D and I, where each site-mean direction is treated as a vector of unit length; α_{95} is the radius of the circle of 95% confidence; PLONG, PLAT are mean longitude and latitude of the Goble paleomagnetic pole, calculated using site virtual geomagnetic poles as data. For a discussion of paleomagnetic statistical techniques, see Irving (1965). Without tilt correction the mean direction for the Goble Volcanic Series is 18.5°, 53.5°, 4.5° (D, I, α_{95}). Data for middle Tertiary pole: Grommé and McKee (1971), Gose (1970), Gilliland and others (1969), Lovlie and Opdyke (1974), Currie and Larochelle (1969), Beck and others (1977), Shive and Pruss (1977). Mistastin Lake volcanic rocks (Currie and Larochelle, 1969), formerly considered Triassic in age, are now assigned to the Tertiary. See discussion in Irving and others (1976).

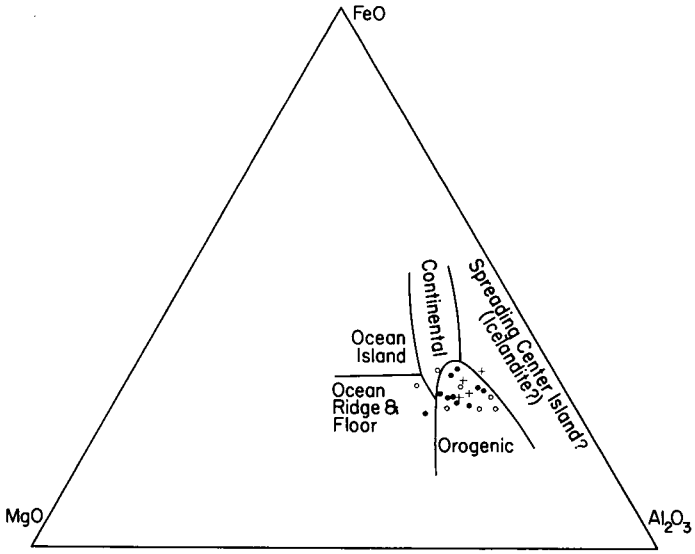


Figure 4. Geochemical discriminant diagram, after Pearce and others (1977). Filled circles = Goble Volcanic Series; open circles = volcanic rock units in Washington Cascades; plus symbol = volcanic rock units in Oregon Cascades. Data from Peck and others (1964), Swanson (1966), Wise (1970), Burr (1978), and Snavely and MacLeod (1974). FeO is total iron calculated as FeO.

the alternating field technique. Magnetic stability was high for the most part, although four sites failed to give circles of 95% confidence of less than 15° and were rejected. Mean magnetic directions for the remaining 38 sites form a tight group except for site 77-20B, which has a strong westerly declination, a steep inclination, and a petrological character suggesting that it may be much younger than the other volcanic exposures in the area (P. Hammond, 1978, personal commun.). Therefore, site 77-20B also was rejected; this left 333 samples from 37 sites with which to determine the mean direction of remanent magnetization of the Goble Volcanic Series. Of the 37 sites, 28 were of normal magnetic polarity and 9 were reverse. Figure 2 shows the distribution of site-mean directions, after magnetic cleaning and correction for tilt.

The question of whether or not to correct for tilt proved to be one of the most intractable aspects of the investigation. Several broad, open folds that trend northwest through the region have been mapped; these folds are transverse up to the north-south axis of the Coast Range and Cascades (see, for example, Henriksen, 1956; Roberts, 1958). The several steeply dipping flows seen in the Goble Volcanic Series thus almost certainly show postmagnetization tilting; therefore, their mean directions of magnetization must be corrected for tilt in order to recover the original direction of the geomagnetic field. Thus, when all flows (a total of 10) with dips steeper than 10° were corrected for dip, a considerable reduction in scatter was apparent (Burr, 1978, Fig. 7). However, many of the slight dips measured in outcrop very likely represent original dip or instances of flows engulfing irregular topography. Correcting for such "dips" will, of course, introduce error into the calculation of mean direction for the Goble Volcanic Series. The problem of unerringly distinguishing tectonic from original dips was one we were unable to solve, and in the last analysis we corrected all exposures for dip. The scatter in mean directions for the volcanic series was reduced thereby, but not significantly (the precision parameter of Fisher [1953] increased after dip correction from 27.45° to 30.54°). Because tilts, however, are small and randomly directed, whether or not we corrected for tilt makes no difference in the discussion that follows; mean directions for the Goble Volcanic Series, calculated with and without tilt corrections, differ by only 5° and are not statistically distinct. Table 1 gives the mean direction of remanent magnetization for the Goble Volcanic Series, after magnetic cleaning and correction for tilt. Included for comparison is the expected direction for middle Tertiary rock units in southern Washington calculated from the mean pole for stable North America. It is evident from Table 1 that the mean declination of the Goble rocks is rotated sharply eastward; this implies that the Goble block, like the rocks of the Oregon Coast Range, has been rotated clockwise relative to the stable interior of the continent. The amount of rotation of the Goble block is 25° ± 13° (95% confidence limits; Burr, 1978, Fig. 10).

The rotation of the Goble block implied by its paleomagnetism is in the same sense as the rotation of the Oregon Coast Range, but in absolute amount it is considerably smaller. Is the difference in rotation-angle meaningful? We suspect that it is, for the following reasons:

1. Although the circles of confidence for the Goble Volcanic Series and the upper Eocene Yachats basalts (Simpson and Cox, 1977) of the Coast Range overlap slightly, as shown in Figure 3, the circle of confidence for the Yachats basalts is extraordinarily large, as it is based on only eight flows, which are all that are exposed in the Yachats outcrop area. The mean direction for Oligocene sills and dikes in the Coast Range shows less rotation than the direction for the Yachats flows (P. J. Plumley, 1978, personal commun., not shown in Fig. 3), as one would expect if the Coast Range were rotating during the late Eocene and Oligocene; however, it too shows more displacement than the

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direction for the Goble Volcanic Series. Work in progress (J. Magill and A. Cox, 1978, personal commun.) on the Tillamook Volcanic Series may resolve the question of whether or not the mean direction of the Goble Volcanic Series is statistically distinct from directions of rock units of equivalent age in the Oregon Coast Range. Whether or not the Goble block is part of the Coast Range block may still be in question, however, because sampling problems and difficulties with tilt corrections make it difficult to apply the relevant statistical techniques correctly.

2. Some geochemical evidence suggests that the Goble rocks may have a different origin than the various volcanic complexes of the Oregon Coast Range. The latter rocks, together with basalts of similar age in western Washington (for example, the Crescent Formation) and the Metchosin Volcanic Series of southern Vancouver Island, Canada, have been attributed to an oceanic environment—either ocean-ridge tholeiites or seamounts (Snavely and MacLeod, 1974; Glassley, 1974). As shown in Figure 4, however, the major-element geochemistry of the Goble rocks is more nearly compatible with an orogenic (magmatic arc) environment than with an oceanic derivation. Perhaps the Goble Volcanic Series represents arc tholeiites associated with an early stage in the evolution of the Cascade magmatic arc. The low K content of the Goble rocks (Burr, 1978, Fig. 4b) suggests that they are part of the arc-tholeiite association of Jakeš and White (1972). However, results reported by Arculus and Johnson (1978) cast doubt on the reliability of simple geochemical indicators of the tectonic setting of such volcanic rocks.

3. Perhaps the strongest evidence that the Goble flows are not genetically related to the volcanic rocks of the Coast Range is their present tectonic setting. The Goble flows lie east of the Puget-Willamette lowland and east of a prominent north-trending zone of high gradient in Bouguer gravity anomaly values (Bonini and others, 1974; Berg and Thiruvathukal, 1967). The Oregon Coast Range, which is west of the gradient, has relatively high anomaly values (0 to +60 mgal) that probably reflect the existence of thin, basaltic crust. The gradient itself averages 2 to 4 mgal/km into and through the Puget-Willamette lowland. East of the lowland, anomaly values range from −30 to −100 mgal; they probably reflect thickened crust associated with the Cascade magmatic arc. The Goble Volcanic Series lies in an area where the Bouguer gravity anomaly shows values of −30 to −50 mgal and thus probably is underlain by the thicker crust characteristic of the Cascades. The striking change in Bouguer anomaly values from +30 mgal over typical Coast Range volcanic rocks west of Olympia, Washington, to −80 mgal in the core of the Olympic Mountains suggests to us that the Olympics and the Coast Range have radically different crustal thicknesses and compositions and thus also are not part of the same crustal block.

DISCUSSION

The Coast Range block must have rotated between about 50 and 25 m.y. B.P.; data available to us do not preclude some rotation before or after this time but do not require it. Rotation of the Goble block must have occurred after extrusion of the Goble flows in the late Eocene or early Oligocene and apparently was completed before the eruption of the Frenchman Springs basalts (dated at 14 to 15 m.y. B.P.; Kienle and others, 1978), which crop out near Goble exposures in northwest Oregon. During this period the continental margin was being underthrust by the Farallon plate (Atwater, 1970). Subduction was rapid and oblique, with a significant component of right-lateral shear (Carlson, 1976; Coney, 1978; M. E. Beck, Jr., unpub. data). Because inclinations of remanent magnetization found in Coast Range (and Goble) rocks are not anomalous (that is, they are not significantly different from values calculated from the mean middle Tertiary pole for the North American craton), it appears that these crustal blocks did not move northward relative to interior North America. The rapid change in magnetic declination with time found in Coast Range rocks, on the other hand, indicates that they rotated about a pivot that lay close at hand. Thus, the Coast Range block appears to have rotated substantially in place. Although the notorious inability of paleomagnetism to detect motion along circles of paleolatitude introduces uncertainty into a purely paleomagnetic reconstruction, evidence summarized by Simpson (1977) seems to require the southern end of the Coast Range block to have been close to, or in contact with, older continental rocks at the time of the middle Eocene deposition of the Tyee and Flournoy sediments (Fig. 1).

Simpson and Cox (1977) have advanced two models to account for paleomagnetic directions found in the Coast Range rocks. In their first model, the Coast Range block, not including the Olympic Mountains, rotates about a southern pivot point near its juncture with the Klamath Mountains in southwestern Oregon. In their second model, the block (including the Olympic Mountains) first moved northeastward as part of the Farallon plate and choked an early Tertiary subduction zone that may have extended southeastward from the Olympic Mountains approximately parallel to the Olympic-Wallowa lineament (Dickinson, 1976; Snyder and others, 1976). Following its arrival at the trench, subduction is thought to have jumped to the seaward edge of the Coast Range block, which then rotated into its present orientation about a pivot point in the Olympic Mountains. Both of these models have attendant geologic difficulties, which are summarized by Simpson (1977).

Our new data seem to complicate, rather than clarify, the issue. The Goble block almost certainly has rotated. If it is part of the Cascade magmatic arc, then at least part of the arc also has rotated in the same sense as the Coast Range and perhaps by nearly as much. Two basic models are possible: (1) Perhaps Simpson and Cox's (1977) model 1 is correct, and the rotation of the Goble block represents an independent response to oblique subduction of the Farallon plate before or during the arrival and accretion of the Coast Range block. (2) Alternatively, following Simpson and Cox's (1977) model 2, perhaps the same process that caused the Coast Range to pivot clockwise after it choked the early Tertiary trench also carried the magmatic arc to its present location. Both models imply that arc rocks younger than Goble, but perhaps older than late Miocene, also will show rotation. The only paleomagnetic study of pre-Miocene Cascade volcanic rocks to our knowledge is Beck (1962), which deals with Eocene to Miocene flows exposed in the western flank of the Cascades along the northern California border. These rocks also show an eastward-directed declination but with a wide confidence interval. Work in the planning stage on arc rocks of late Eocene and Oligocene age in Washington and Oregon should help us decide whether the early Cascade magmatic arc rotated as part of the Coast Range crustal block, or independently as a mosaic of smaller blocks, or has had some other, entirely unexpected, history.

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